Garrett Fincke

Pittsburgh, PA | garrettfincke@gmail.com | fincke.dev | github.com/ggfincke | linkedin.com/in/garrett-fincke

Education

University of Pittsburgh Aug 2026 – Expected May 2028
Master of Science in Computer Science; SCI Scholarship for Excellence *Pittsburgh, PA*

The Pennsylvania State University Aug 2021 – Dec 2024
Bachelor of Science in Computer Science *University Park, PA*

Experience

Westinghouse Electric Company May 2026 – Dec 2026
Software Developer Intern, GIC Tools and Applications *Cranberry Township, PA*

- Built and deployed a Python/FastAPI service over legacy Oracle document data with audit-trailed write/undo, corporate auth, project access controls, rate limiting, request-ID logging, and health probes
- Validated deterministic retrieval across 500+ releases, 1,700+ hierarchy locations, and 60,000 CI memberships
- Leading backend development of DocMon to take a human out of the loop on tedious cross-system revision reconciliation; gathered stakeholder requirements and built its reconciliation engine, parallelizing throttled API calls across 16 workers for 12x faster comparisons
- Integrating the document API with Westinghouse's Databricks agent platform to support bertha generative AI workflows; connecting it to an internal project database to automate stale-revision checks

Scale AI May 2024 – Jul 2025
Software Engineer (contract) *Remote*

- Built Python and Docker evaluation harnesses for LLM-generated code, automating compilation, execution, linting, testing, pass@k scoring, and failure analysis
- Created schema-versioned Pydantic and JSON Schema pipelines with deterministic sampling, replacing ad hoc manual review with reproducible validation and reducing evaluation turnaround time
- Built internal dashboards for evaluation failure analysis and PostgreSQL ingest checks for schema normalization and deduplication, improving data quality across annotation and QA workflows

Projects

SwimMate v2 | *Python, Django REST, PostgreSQL/pgvector, OpenAI, Swift, SwiftUI, iOS/watchOS*
Full-stack swimming platform with native iOS/watchOS apps and AI-generated workouts

- Built a Django/PostgreSQL RAG pipeline for personalized swim workouts with hybrid retrieval and reranking; added a deterministic 48-case eval suite tracking constraint pass rates, latency, and token usage
- Built SwiftUI iOS/watchOS apps with HealthKit workouts, offline sync, token refresh, and WatchConnectivity

Coral | *TypeScript, Node.js, Ollama, MCP, ACP, SQLite, React Ink*
Local-first Ollama coding agent with research subagents, MCP/ACP integration, and durable tasks

- Built a multi-turn agent loop with parallel tool use across 19 tools and read-only research subagents, a stdio MCP client with launch trust and schema validation, and an ACP server for editor clients
- Built durable coding jobs running approved work in isolated Git worktrees with checkpoints and crash recovery
- Hardened local-model reliability with tool-call recovery, schema validation, self-verification, and loop detection

MDX Preview for VS Code | *TypeScript, React 19, MDX 3, VS Code Extension API*
Published VS Code/Open VSX extension for live MDX previews with 15K+ combined installs

- Architected a six-package monorepo and published its reusable compiler/runtime as the mdx-forge npm package
- Built trust-gated Safe and Trusted rendering with DOMPurify, strict CSP, React component support, and compatibility shims for five MDX frameworks

Technical Skills

Languages: Python, TypeScript/JavaScript, Swift, Java, SQL, C/C++
Backend & Data: Django, FastAPI, Node.js, PostgreSQL/pgvector, Oracle, SQLite, Pydantic, REST/OpenAPI
AI/LLM: RAG, embeddings/vector search, agentic AI, MCP, Ollama, OpenAI/Anthropic APIs
Frontend & Mobile: React, Next.js, SwiftUI, UIKit, Tailwind CSS
Cloud & DevOps: Docker, Docker Compose, AWS EC2/SSM/CloudWatch, GitHub Actions, CI/CD, Vercel